



Homework 3

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Problem 3.1

Proof.

$$S(n) \equiv \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \forall n \in \mathbb{N}, n \geq 1$$

Let us define $C(n)$ as

$$C(n) = \left\{ n \in \mathbb{N} \mid \sum_{k=1}^n k^2 \neq \frac{n(n+1)(2n+1)}{6} \right\}$$

We assume $C(n)$ is not null so there should exist a least element n .

We check $n = 1$

$$\left\{ \begin{array}{l} \sum_{k=1}^1 k^2 = 1^2 = 1 \\ \frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{6}{6} = 1 \end{array} \right. \implies 1 \notin C(n) \implies n > 1$$

There should exist a least element $n > 1$ in $C(n)$ so $n - 1 \notin C(n)$

We prove that for some $n \in \mathbb{N}^*$ the $S(n - 1) + n^2 = S(n)$ does not hold

$$\begin{aligned} \left\{ \begin{array}{l} S(n - 1) + n^2 = \sum_{k=1}^{n-1} k^2 = \frac{(n-1)n(2(n-1)+1)}{6} + n^2 = \frac{2n^3 - 3n^2 + n}{6} + n^2 = \frac{2n^3 + 3n^2 + n}{6} \\ S(n) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6} \end{array} \right. \implies \\ \implies S(n - 1) + n^2 = S(n) \end{aligned}$$

So it holds for all $n \in \mathbb{N}^*$ which means $C = \emptyset$ and by contradiction $S(n)$ holds for all $n \in \mathbb{N}^*$ $\square$

Problem 3.2

- a) *Proof.* Suppose, for contradiction, that the store owner is wrong, so \$12 can be made from \$5 and \$8 cards. Define

$$S = \{ b \in \mathbb{N} \mid \exists a \in \mathbb{N} \text{ such that } 5a + 8b = 12 \}.$$

This is the set of possible numbers of \$8 cards used in some way of making \$12. By assumption S is nonempty, and $S \subseteq \mathbb{N}$. By the Well Ordering Principle, S has a least element; call it b_0 .

Because $b_0 \in S$, there exists $a_0 \in \mathbb{N}$ with

$$5a_0 + 8b_0 = 12.$$

We know that $5a_0 \geq 0$, so $8b_0 \leq 12$ and $b_0 \leq 1$. Hence $b_0 = 0$ or $b_0 = 1$.

- If $b_0 = 0$, then $5a_0 = 12$, so $a_0 = \frac{12}{5} \notin \mathbb{N}$, a contradiction.
- If $b_0 = 1$, then $5a_0 = 12 - 8 = 4$, so $a_0 = \frac{4}{5} \notin \mathbb{N}$, a contradiction.

Both cases are impossible, so S must be empty, contradicting our assumption. Therefore \$12 cannot be made using only \$5 and \$8 cards, and the store owner is correct. $\square$



b) *Proof.* Suppose, for contradiction, that some integer $n \geq 28$ is not formable. Then the set

$$C = \{n \in \mathbb{N} : n \geq 28 \text{ and } n \text{ is not formable}\}$$

is a nonempty set of natural numbers. By the Well Ordering Principle, C has a least element m . Since $m \geq 28$, we have two cases.

Case 1: $28 \leq m \leq 32$. All elements are formable:

$$28 = 4 \cdot 5 + 1 \cdot 8,$$

$$29 = 1 \cdot 5 + 3 \cdot 8,$$

$$30 = 6 \cdot 5 + 0 \cdot 8,$$

$$31 = 3 \cdot 5 + 2 \cdot 8,$$

$$32 = 0 \cdot 5 + 4 \cdot 8.$$

Case 2: $m \geq 33$. Then $m - 5 \geq 28$. We can write $m - 5 = 5a + 8b$ $a, b \geq 0$. Then

$$m = 5(a + 1) + 8b,$$

so m is formable, contradicting $m \in C$.

Both cases lead to a contradiction, so C is empty. Therefore every amount $n \geq 28$ can be formed using \$5 and \$8 cards. $\square$

Problem 3.3

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}, \quad \forall n \in \mathbb{N}, n > 1.$$

Proof. Lets us define: $f(n) = \sum_{i=1}^n \frac{1}{i^2}$.

1. Base case ($n = 2$): For $n = 2$, if we evaluate both sides we get:

$$\text{LHS} = f(2) = 1 + \frac{1}{4} = \frac{5}{4} = 1.25, \quad \text{RHS} = 2 - \frac{1}{2} = \frac{3}{2} = 1.5.$$

Since $1.25 < 1.5$, the inequality holds true.

2. We prove the induction

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{k^2} < 2 - \frac{1}{k}.$$

$$1 + \frac{1}{4} + \cdots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \frac{1}{(k+1)^2}.$$

$$2 - \frac{1}{k} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1} \iff \frac{1}{(k+1)^2} \leq \frac{1}{k} - \frac{1}{k+1}.$$

$$\frac{1}{k} - \frac{1}{k+1} = \frac{(k+1) - k}{k(k+1)} = \frac{1}{k(k+1)}.$$



$$\frac{1}{(k+1)^2} < \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

$$1 + \frac{1}{4} + \cdots + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \left(\frac{1}{k} - \frac{1}{k+1} \right) = 2 - \frac{1}{k+1}.$$

□

Problem 3.4

$$F_1 + F_2 + F_3 + \cdots + F_n = F_{n+2} - 1, \quad \forall n \in \mathbb{N}, n \geq 1.$$

Proof. From the definition of the Fibonacci sequence: $F_1 = 1$, $F_2 = 1$, and $F_k = F_{k-1} + F_{k-2}$ for all integers $k \geq 3$.

We define be the statement: $f(n) = \sum_{i=1}^n F_i = F_{n+2} - 1$.

1. Base Case ($n = 1$): For $n = 1$, if we evaluate both sides:

$$\text{LHS} = f(1) = F_1 = 1, \quad \text{RHS} = F_{1+2} - 1 = F_3 - 1 = 2 - 1 = 1.$$

Since LHS = RHS, the inequality holds true.

2. We prove the induction

$$F_1 + F_2 + F_3 + \cdots + F_k = F_{k+2} - 1.$$

$$F_1 + F_2 + \cdots + F_k + F_{k+1} = F_{(k+1)+2} - 1 = F_{k+3} - 1.$$

$$\begin{aligned} \sum_{i=1}^{k+1} F_i &= \left(\sum_{i=1}^k F_i \right) + F_{k+1} \\ &= (F_{k+2} - 1) + F_{k+1} \\ &= (F_{k+2} + F_{k+1}) - 1. \end{aligned}$$

We can rewrite F_{k+3} as, $F_{k+3} = F_{k+2} + F_{k+1}$. Substituting this in:

$$\sum_{i=1}^{k+1} F_i = F_{k+3} - 1.$$

Thus the equality holds true.

□



Problem 3.5

Proof.

$$4^n + 7 \leq 5^n$$

1. Base Case ($n = 2$): For $n = 2$, we evaluate both sides:

$$\text{LHS} = 4^2 + 7 = 16 + 7 = 23, \quad \text{RHS} = 5^2 = 25.$$

Since $23 \leq 25$, the equality holds true.

2. We prove the induction

$$4^k + 7 \leq 5^k.$$

$$\begin{aligned} 4^{k+1} + 7 &= 4 \cdot 4^k + 7 \\ &< 5 \cdot 4^k + 35 \quad (\text{since } 4 \cdot 4^k < 5 \cdot 4^k \text{ and } 7 < 35) \\ &= 5(4^k + 7). \end{aligned}$$

$$5(4^k + 7) \leq 5 \cdot (5^k) = 5^{k+1}.$$

$$4^{k+1} + 7 \leq 5^{k+1}.$$

Thus, the inequality holds. □

Problem 3.6

A positive integer n is called a **perfect number** if it is equal to the sum of its proper positive divisors. Prove by induction that if $2^k - 1$ is a prime number (a **Mersenne prime**), then $n = 2^{k-1}(2^k - 1)$ is an even perfect number.

Proof. Let $p = 2^k - 1$ be a Mersenne prime. Since p is prime and $p > 1$, we must have $k \geq 2$, so $n = 2^{k-1}p$ is divisible by 2 and is therefore even.

The proper positive divisors of $n = 2^{k-1}p$ consist of:

1. Divisors that are powers of 2: $1, 2, 2^2, \dots, 2^{k-1}$.
2. Divisors multiplied by p (excluding n itself): $p, 2p, 2^2p, \dots, 2^{k-2}p$.

Thus, the sum of all proper positive divisors of n is:

$$S = \sum_{i=0}^{k-1} 2^i + p \sum_{i=0}^{k-2} 2^i.$$

We first prove by induction that for any integer $m \geq 0$:

$$g(m) = \sum_{i=0}^m 2^i = 2^{m+1} - 1.$$



1. Base Case ($m = 0$): For $m = 0$:

$$\text{LHS} = g(0) = 2^0 = 1, \quad \text{RHS} = 2^{0+1} - 1 = 2 - 1 = 1.$$

2. We prove the induction

$$\sum_{i=0}^{j+1} 2^i = \left(\sum_{i=0}^j 2^i \right) + 2^{j+1} = (2^{j+1} - 1) + 2^{j+1} = 2 \cdot 2^{j+1} - 1 = 2^{j+2} - 1.$$

Thus, the sum formula holds for all $m \geq 0$.

Using $g(k-1)$ and $g(k-2)$:

$$\sum_{i=0}^{k-1} 2^i = 2^k - 1 = p, \quad \sum_{i=0}^{k-2} 2^i = 2^{k-1} - 1.$$

$$S = p + p(2^{k-1} - 1) = p(1 + 2^{k-1} - 1) = p \cdot 2^{k-1} = n.$$

Since the sum of the proper positive divisors of n equals n , n is a perfect number. Since $k \geq 2$, n is an even perfect number. $\square$

Problem 3.7

Proof.

$$F_n < 2^n.$$

1. Base Cases ($n = 1$ and $n = 2$): For $n = 1$:

$$\text{LHS} = F_1 = 1, \quad \text{RHS} = 2^1 = 2.$$

Since $1 < 2$, the inequality holds true.

For $n = 2$:

$$\text{LHS} = F_2 = 1, \quad \text{RHS} = 2^2 = 4.$$

Since $1 < 4$, the inequality holds true.

2. We prove the induction Assume by strong induction that $F_i < 2^i$ holds for all $1 \leq i \leq k$, where $k \geq 2$. We show that $F_{k+1} < 2^{k+1}$.

By definition of the Fibonacci sequence:

$$F_{k+1} = F_k + F_{k-1}.$$

Applying the strong inductive hypothesis to F_k and F_{k-1} (since $1 \leq k-1 < k$):

$$F_k < 2^k \quad \text{and} \quad F_{k-1} < 2^{k-1}.$$

Substituting these into the expression for F_{k+1} :

$$F_{k+1} = F_k + F_{k-1} < 2^k + 2^{k-1}.$$



Since $2^{k-1} < 2^k$ for all $k \geq 2$:

$$2^k + 2^{k-1} < 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}.$$

Therefore:

$$F_{k+1} < 2^{k+1}.$$

Thus, the inequality holds. □

Problem 3.8

Proof. Let $P(n)$ be the statement that an amount of \$ can be formed using natural number quantities of 5\$ and 8\$ cards: $n = 5a + 8b$ for some $a, b \in \mathbb{N}$.

1. Base Cases ($n = 28, 29, 30, 31, 32$): We evaluate $P(n)$ for five consecutive base values:

- For $n = 28$: $28 = 5(4) + 8(1)$.
- For $n = 29$: $29 = 5(1) + 8(3)$.
- For $n = 30$: $30 = 5(6) + 8(0)$.
- For $n = 31$: $31 = 5(3) + 8(2)$.
- For $n = 32$: $32 = 5(0) + 8(4)$.

Because all are valide the base case is true.

2. We prove the induction We assume by strong induction that $P(i)$ is true for all $28 \leq i \leq k$, where $k \geq 32$. We prove that $P(k+1)$ is true.

$$(k+1) - 5 = k - 4 \geq 28.$$

Since $28 \leq k - 4 \leq k$, by the strong inductive hypothesis $P(k-4)$ holds true, so there exist $a', b' \in \mathbb{N}$ such that:

$$k - 4 = 5a' + 8b'.$$

Adding \$5 to both sides yields:

$$k + 1 = (k - 4) + 5 = (5a' + 8b') + 5 = 5(a' + 1) + 8b'.$$

Since $a' \geq 0 \implies a' + 1 \in \mathbb{N}$ and $b' \in \mathbb{N}$, this provides a valid construction for $$(k+1)$ using $5 and $8 cards.$

Thus, $P(k+1)$ his true. □



Problem 3.9

Flaw in the Proof:

The flaw in the bogus proof occurs in the **inductive step** when $n = 0$.

Specifically, the proof relies on the algebraic identity:

$$a^{n+1} = \frac{a^n \cdot a^n}{a^{n-1}}$$

and substitutes $a^n = 1$ and $a^{n-1} = 1$ using the induction hypothesis $P(n) : \forall k \leq n, a^k = 1$.

1. Invalid Exponent for $n = 0$: When $n = 0$, the expression requires $a^{n-1} = a^{-1}$:

$$a^{0+1} = a^1 = \frac{a^0 \cdot a^0}{a^{-1}}$$

The induction hypothesis $P(0)$ only asserts that $a^k = 1$ for non-negative integers $k \leq 0$ (i.e., $k = 0$, giving $a^0 = 1$). It provides no information about $k = -1$, because -1 is not a non-negative integer ($k \notin \mathbb{N}$). Thus, substituting $a^{-1} = 1$ is invalid.

2. Breakdown of the Inductive Chain: For $a^{n-1} = 1$ to follow legitimately from the induction hypothesis $P(n)$, $n - 1$ must be a non-negative integer, which requires $n \geq 1$. Consequently, $P(0)$ cannot imply $P(1)$. Since the transition from $n = 0$ to $n = 1$ fails, the entire inductive argument collapses.

Problem 3.10

Proof. Let (x, y) denote Romeo's coordinates on the grid. We define the invariant quantity:

$$I(x, y) = x + 2y \pmod{7}.$$

1. Initial State: At Romeo's starting position $(0, 0)$, we evaluate the invariant:

$$I(0, 0) = 0 + 2(0) = 0 \equiv 0 \pmod{7}.$$

2. Invariance under moves: Suppose Romeo is at position (x, y) such that $x + 2y \equiv 0 \pmod{7}$. Let (x', y') be his new position after applying any of the four allowed moves:

1. Move 1 $(+2, -1)$: $x' + 2y' = (x + 2) + 2(y - 1) = x + 2y + 0 \equiv 0 \pmod{7}$.
2. Move 2 $(-2, +1)$: $x' + 2y' = (x - 2) + 2(y + 1) = x + 2y + 0 \equiv 0 \pmod{7}$.
3. Move 3 $(+1, +3)$: $x' + 2y' = (x + 1) + 2(y + 3) = x + 2y + 7 \equiv 0 \pmod{7}$.
4. Move 4 $(-1, -3)$: $x' + 2y' = (x - 1) + 2(y - 3) = x + 2y - 7 \equiv 0 \pmod{7}$.

In every case, $x' + 2y' \equiv 0 \pmod{7}$. By induction on the number of moves, any position reachable by Romeo must satisfy $x + 2y \equiv 0 \pmod{7}$.



3. Target Evaluation: For Julietta's position at $(1, 1)$, we have:

$$1 + 2(1) = 3 \not\equiv 0 \pmod{7}.$$

Since $3 \not\equiv 0 \pmod{7}$, the point $(1, 1)$ does not satisfy the invariant. Therefore, Romeo will never be able to reach Julietta. $\square$

Problem 3.11

Proof. We write (B, G, K) for the numbers of blue, green, and black chameleons, respectively, so the initial state is $(11, 13, 17)$.

1. Invariant. In any meeting of two chameleons of different colors, the two colors that meet each decrease by 1 and the third color increases by 2. For example,

$$\text{Blue \& Green meet: } (B, G, K) \mapsto (B - 1, G - 1, K + 2).$$

Thus each of B, G, K changes by either -1 or $+2$ in every meeting. Since $-1 \equiv 2 \pmod{3}$, all three counts change by the same residue modulo 3. Therefore each of the differences

$$B - G, \quad G - K, \quad K - B$$

is unchanged modulo 3 by every meeting. Explicitly, in the example above,

$$\begin{aligned} (B - 1) - (G - 1) &= B - G, \\ (G - 1) - (K + 2) &= (G - K) - 3 \equiv G - K \pmod{3}, \\ (K + 2) - (B - 1) &= (K - B) + 3 \equiv K - B \pmod{3}, \end{aligned}$$

and the other two types of meeting are identical up to relabeling colors.

2. All blue and all black are impossible. Initially,

$$B - G = 11 - 13 = -2 \equiv 1, \quad G - K = 13 - 17 = -4 \equiv 2, \quad K - B = 17 - 11 = 6 \equiv 0 \pmod{3}.$$

- **All blue**, $(41, 0, 0)$: here $G - K = 0$. By the invariant we would need $G - K \equiv 0 \pmod{3}$ initially, but $G - K \equiv 2 \pmod{3}$. Impossible.
- **All black**, $(0, 0, 41)$: here $B - G = 0$. By the invariant we would need $B - G \equiv 0 \pmod{3}$ initially, but $B - G \equiv 1 \pmod{3}$. Impossible.

3. All green is achievable. The invariant does not rule out $(0, 41, 0)$, since $K - B \equiv 0 \pmod{3}$ initially. In fact it can be reached:

1. Let a black and a green chameleon meet, twice in succession:

$$(11, 13, 17) \longrightarrow (13, 12, 16) \longrightarrow (15, 11, 15).$$

2. Now $B = K = 15$. Let a blue and a black chameleon meet 15 times. Each meeting changes (B, G, K) by $(-1, +2, -1)$, so the result is

$$(15 - 15, 11 + 2 \cdot 15, 15 - 15) = (0, 41, 0).$$

Hence all blue and all black are impossible, while all green is achievable; in particular, the statement that no monochromatic state can be reached is false for the initial state $(11, 13, 17)$. $\square$



Problem 3.12

Proof. We analyze the number of black and white squares on the chessboard.

1. Color Invariant: Color the 8×8 chessboard in the usual alternating pattern of black and white squares. Since the chessboard has an even number of squares in each row and column, opposite corners have the same color.

Initially, the chessboard contains

32 black squares and 32 white squares.

Removing two opposite corners therefore removes two squares of the same color. Without loss of generality, suppose both removed squares are black. The remaining board then contains

30 black squares and 32 white squares.

2. Effect of a Domino: A domino always covers two adjacent squares. Since adjacent squares on a chessboard have opposite colors, every domino covers exactly one black square and one white square.

Therefore, if 31 dominoes were used, they would cover exactly

31 black squares and 31 white squares.

3. Contradiction: However, after removing the two opposite corners, the chessboard contains 30 black squares and 32 white squares. Thus, the numbers of black and white squares are not equal.

Since every domino covers one square of each color, it is impossible for 31 dominoes to cover the remaining board.

Therefore the chessboard cannot be tiled with 31 dominoes.

□

Problem 3.13

Proof. We define by $n_L(w)$ the number of left brackets and by $n_R(w)$ the number of right brackets in w . We wish to prove that for all $w \in BM$, $n_L(w) = n_R(w)$.

Base Case For $w = \lambda$, $n_L(\lambda) = 0$ and $n_R(\lambda) = 0$. Since $0 = 0$, $n_L(\lambda) = n_R(\lambda)$ holds.

We prove the induction Let $w \in BM$ be generated by the constructor rule, so $w = [s]t$ for some $s, t \in BM$.

We assume $n_L(s) = n_R(s)$ and $n_L(t) = n_R(t)$.



By counting brackets in $w = [s]t$:

$$n_L(w) = 1 + n_L(s) + n_L(t)$$

$$n_R(w) = 1 + n_R(s) + n_R(t)$$

Applying the induction hypothesis $n_L(s) = n_R(s)$ and $n_L(t) = n_R(t)$:

$$n_L(w) = 1 + n_L(s) + n_L(t) = 1 + n_R(s) + n_R(t) = n_R(w).$$

Thus, $n_L(w) = n_R(w)$ holds for $w = [s]t$. □

Problem 3.14

a) $1/x$

1. $A = x$

2. $B = 2^A = 2^x$

3. $C = B^{-1} = \log_2 x$

4. $D = 2^{-1 \times C} = 2^{-\log_2 x} = (2^{\log_2 x})^{-1} = x^{-1} = \frac{1}{x}$

b) We prove that $\tan(x)$ is in EF, then that $\frac{x-5}{x+5}$ is in EF

1. $A = x$

2. $B = \frac{1}{x}$

3. $C = \sin(x)$

4. $D = AA = xx = x^2$

5. $E = C \circ (x + \frac{\pi}{2}) = \cos x$

6. $F = C \times B \circ E = \sin x \times \frac{1}{\cos x} = \tan x$

7. $G = (A - 5) \times B \circ (A + 5) = \frac{x-5}{x+5}$

8. $H = F \circ G = \tan(\frac{x-5}{x+5})$

c) $\ln(x^2 + 1)$

1. $A = x$

2. $B = \frac{1}{x}$

3. $C = \log_2 x$

4. $D = AA = xx = x^2$ where $e = 2.71 \dots$ and is the euler number

5. $E = C \circ A \times B \circ (C \circ e) = \log_2 x \times \frac{1}{\log_2 e} = \frac{\log_2 x}{\log_2 e} = \ln x$

6. $F = A \times A + 1 = x^2 + 1$

7. $H = E \circ F = \ln(x^2 + 1)$

d) Proof by structural induction that if $f \in EF$, then $f' \in EF$.

1. **Base Cases:**

- i. $f(x) = x \implies f'(x) = 1 \in EF$ (constant)

- ii. $f(x) = c \implies f'(x) = 0 \in EF$ (constant)

- iii. $f(x) = \sin(x) \implies f'(x) = \cos(x) \in EF$ (proven in part b)

2. **Inductive Hypothesis:** WE assume $f, g \in EF$ with $f', g' \in EF$.

3. **Inductive Step:**

- i. $(f + g)' = f' + g' \in EF$ (sum rule)

- ii. $(fg)' = f' \times g + f \times g' \in EF$ (product rule)

- iii. $(2^g)' = 2^g \times g' \times \ln(2) \in EF$ ($\ln(2) = \frac{1}{\log_2 e} \in EF$ because is constant)

- iv. $(f^{-1})' = B \circ f' \circ f^{-1} \in EF$ $B(x) = \frac{1}{x} \in EF$ (proven in part a)



v. $(f \circ g)' = (f' \circ g) \times g' \in EF$ (chain rule)

Vitalie is NS (natural stupidity) and can make mistakes. Please perform two factor verification on the Domashneye Rabota